

Multiple Choice

1. **Answer:** B

Options: A) pure; B) not pure; C) useful; D) useless

Note: Correct option: B - not pure

2. **Answer:** B

Options: A) Attributes are equally important.; B) Attributes are statistically dependent of one another given the class value.; C) Attributes are statistically independent of one another given the class value.; D) Attributes can be nominal or numeric

Note: Correct option: B - Attributes are statistically dependent of one another given the class value.

3. **Answer:** C

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: C - True, False

4. **Answer:** C

Options: A) $P(H, U, P, W) = P(H) * P(W) * P(P) * P(U)$; B) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(W | H, P)$; C) $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(U | H, P)$; D) None of the above

Note: Correct option: C - $P(H, U, P, W) = P(H) * P(W) * P(P | W) * P(U | H, P)$

5. **Answer:** A

Options: A) $p(y|x, w)$; B) $p(y, x)$; C) $p(w|x, w)$; D) None of the above

Note: Correct option: A - $p(y|x, w)$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'It discovers causal relationships'.

Long Form Response

11. **Answer:** `def adjac(ele, sub = []): | def get_coordinates(test_tup): | return (res)`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def count_odd(array_nums): | return count_odd`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key